Detecting Patronising and Condescending Language (PCL)
Natural Language Processing Coursework Report

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March 2026

GitHub/GitLab Repository: https://github.com/hardiv/pcl-detector

1 Exercise 1: Critical Paper Review (Stage 1)

1.1 Q1. Primary Contributions

This paper [3] introduces the task of detecting patronising and condescending language (PCL) in text, particularly in the context of online news articles about vulnerable communities, which differs to traditional harmful language detection due to the usually subtler, well-intentioned and unconscious nature of PCL in this context.

The paper's main contributions include the creation of a novel dataset (Don't Patronise Me) for PCL detection, which consists of 10,637 paragraphs about vulnerable social groups selected from news articles annotated for the presence and type of PCL, both at an overall and a more local text-span level.

The paper also establishes experimental baselines across SVM models, BiLSTM models and fine-tuned transformers such as BERT, RoBERTa and DistilBERT for binary and multi-label classification of PCL.

1.2 Q2. Technical Strengths

The dataset is carefully curated and annotated, and draws from multiple news sources across 20 countries over a span of 8 years. It employs the Kappa Inter-Annotator Agreement (IAA) between two annotators (and a third mediating annotator) to give way to a 5-point scale for labels to address the subjective nature of identifying PCL.

The experimental setup is sound, with appropriate train/dev/test splits and evaluation metrics.

The baseline RoBERTa model is well-chosen and provides a reasonable starting point for comparison. With an F_1 score of 70.63 on Task 1, it demonstrates that PCL can be detected, despite its subtlety.

The paper does also identify areas for improvement, such as model failure modes where knowledge gaps due to world knowledge or commonsense reasoning are required, and the need for better handling of class imbalance and more fine-grained PCL types, which can guide future work in this area.

1.3 Weaknesses and Limitations

Whilst care has been taken in the annotation process to avoid subjectivity, around 14% of the labels were disagreed upon by the two annotators, which suggests that the task is inherently subjective and may require more annotators, as biases and annotation inconsistencies could affect model performance and generalisability.

In addition, multi-label performance is diminished by severe category imbalance; the label ünbalanced power relationshäs 1,307 spans versus only 64 for the poorer, the merrier; which contributes to near-zero performance on underrepresented categories. The authors propose using category labels as privileged information but no ablation or empirical validation of this claim is provided.

The paper goes on to suggest that BERT-large has overfitted on the small training dataset, leading it to underperform BERT-base and BiLSTM in Task 1; this suggestion lacks investigation of regularisation or alternative fine-tuning strategies to address overfitting.

The paper also lacks formal interpretability analysis, such as SHAP and LIME, to verify whether the model learns meaningful patterns or is relying on spurious correlations, which would be useful for understanding model behaviour and guiding future improvements.

2 Exercise 2: Exploratory Data Analysis (Stage 2)

2.1 EDA Setup

All analysis was performed in the Jupyter notebook (`main.ipynb`) found in the root of the repository. The labelled corpus comprises the SemEval-2022 train and dev splits of the Don't Patronise Me dataset (10,469 paragraphs in total). Paragraphs with a graded label ≥ 2 are treated as PCL-positive; those with labels 0–1 are PCL-negative. No tokeniser-specific preprocessing was applied during EDA; a simple lower-cased alphabetic-token regex was used for vocabulary and n-gram statistics to remain model-agnostic.

2.2 Technique 1: Statistical Profiling

2.2.1 Visual/Tabular Evidence

Table 1 summarises vocabulary size, token count, and type-token ratio (TTR) for each class. Figure 1 shows the distribution of paragraph lengths in words, annotated with the 95th and 99th percentiles.

Class	Paragraphs	Vocab	Total tokens	TTR
No PCL	9,476	27,572	404,702	0.068
PCL	993	7,627	47,615	0.160
Overall	10,469	29,046	452,317	0.064

Table 1: Vocabulary and token statistics per class.

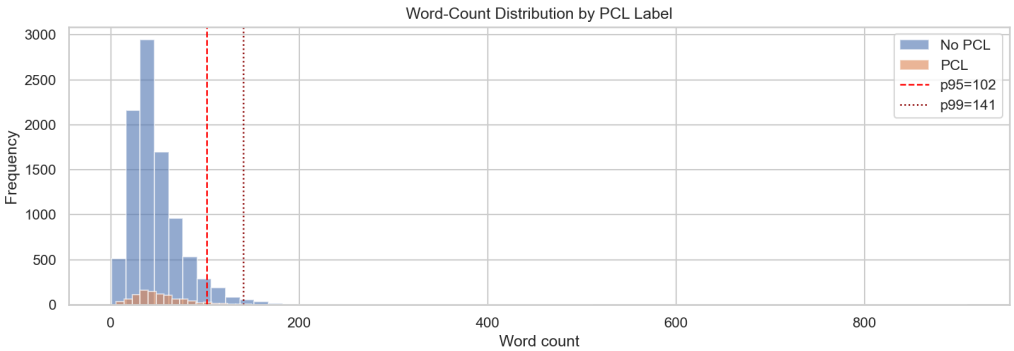

Figure 1: Word-count distribution per class with p95 and p99 cut-offs.

2.2.2 Findings

The dataset is heavily class-imbalanced: 91% of labelled paragraphs are non-PCL, yielding a No-PCL : PCL ratio of roughly 9.5 : 1 (PCL accounts for ∼9.5% of the corpus). The PCL class has a notably higher TTR (0.160 vs. 0.068), indicating greater lexical diversity per token, consistent with the more varied rhetorical strategies used to express condescending language. Furthermore, this PCL rate is not uniform across communities: an analysis of PCL density reveals that 'in-need' and 'homeless' have much higher PCL span counts than 'immigrant', for example.

Paragraph lengths are right-skewed: the median is around 45 words, but the 95th and 99th percentiles reach approximately 102 and 141 words respectively.

A scan found no exact text duplicates and no train-dev leakage in the official splits. Noise artefacts (HTML entities, non-ASCII characters, consecutive spaces) are present in a small fraction of paragraphs (<2%), with no URLs detected.

2.2.3 Impact on Modelling

Class imbalance. A naïve classifier predicting "No PCL" always would achieve ∼91% accuracy, making accuracy a misleading metric. Binary F_1 on the PCL class must be the primary evaluation criterion, and inverse-frequency class weights (or focal loss) should be applied during training to avoid the model collapsing to the majority class.
Sequence length. The p95 of 102 words maps to roughly 130–140 subword tokens for RoBERTa; the p99 of 141 words maps to ∼185 tokens. Setting `max_length=256` therefore covers the entire practical length distribution with comfortable headroom, while remaining half the cost of 512. A safe default is `max_length=256` with truncation.
Noise. The low artefact rate means aggressive cleaning is unnecessary and risks removing meaningful content; simple whitespace normalisation is sufficient.

2.3 Technique 2: Lexical Analysis (N-gram Frequency & Log-Odds)

2.3.1 Visual/Tabular Evidence

Figures 2 and 3 show the top-20 unigrams and bigrams per class after English stop-word removal. Figure 4 shows the 15 unigrams most enriched in each class by log-odds ratio.

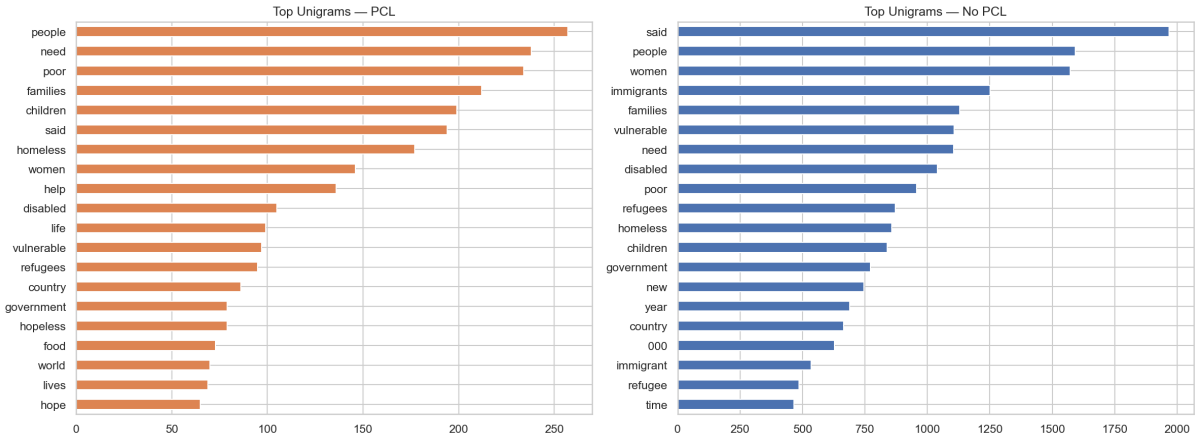

Figure 2: Top-20 unigrams in PCL (orange) and No-PCL (blue) paragraphs.

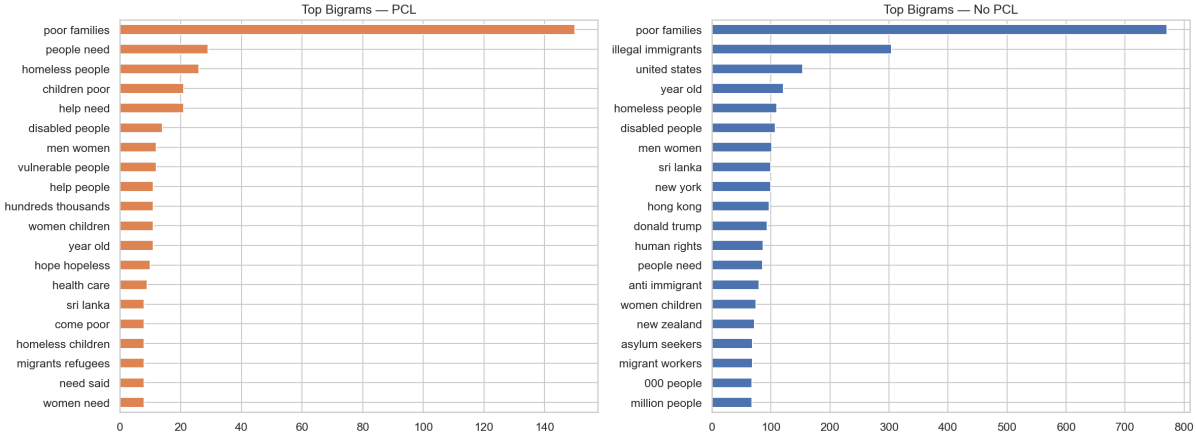

Figure 3: Top-20 bigrams in PCL (orange) and No-PCL (blue) paragraphs.

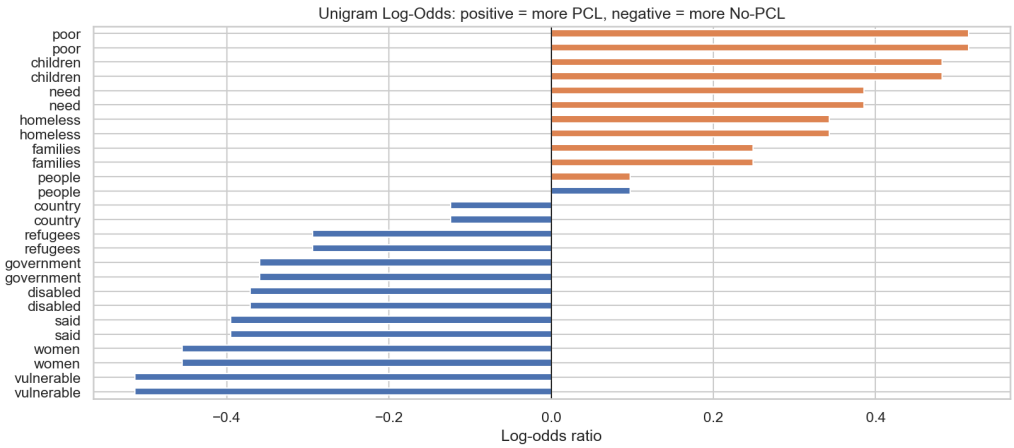

Figure 4: Per-class unigram log-odds ratio; positive values indicate over-representation in PCL, negative in No-PCL.

2.3.2 Findings

Both classes share many high-frequency words (*people, said, year*), confirming that surface term frequency alone gives little signal. The log-odds analysis is more informative: PCL paragraphs are enriched with emotionally loaded and othering terms (*little, help, poor, children, lives, hope*) and phrases like "living in poverty" and "the poorest", directly reflecting the dataset's PCL categories of *poverty* and *over-generalisation*. No-PCL paragraphs are enriched with neutral reporting language (*government, health, percent, new*). The overlap in the most frequent terms illustrates that PCL is driven by framing and semantic context rather than lexical presence alone.

Critically, the n-gram distribution is not uniform across search keywords: communities such as *homeless* and *in-need* contribute disproportionately many high-PCL paragraphs, while keywords like *immigrant* are predominantly non-PCL. This creates a direct risk of label leakage via keyword identity: a model trained without controlling for keyword distribution could learn to predict "PCL" simply because the keyword *homeless* appears, rather than due to any genuine condescension in the text.

2.3.3 Impact on Modelling

No keyword shortcuts. Because discriminative words (*poor, help, children*) also appear frequently in non-patronising contexts, a bag-of-words classifier will struggle. This motivates using a contextual model (e.g., RoBERTa) that encodes the surrounding sentence rather than isolated terms.

Keyword-stratified splitting. The uneven PCL rate across communities poses a concrete risk of label leakage. To prevent the model learning community-specific shortcuts rather than semantic condescension, the internal validation split should be stratified on keyword in addition to label. This ensures every keyword appears in both train and internal-dev folds, forcing the model to generalise across communities rather than memorising the PCL rate of a particular group.

Sequence length. Combining the finding from Technique 1 with the contextual nature of PCL, `max_length=256` is adopted. The p99 of 141 words (∼185 RoBERTa subword tokens) fits well within this budget, ensuring no paragraph in the labelled corpus is truncated in practice, while keeping memory cost well below the 512-token alternative.

3 Exercise 3: Baseline and Proposed Approach (Stage 3)

3.1 Proposed Approach

The baseline fine-tunes RoBERTa-base [2] with standard cross-entropy and no imbalance handling (F_1=0.48). Our model makes five changes.

(1) DeBERTa-v3-base encoder [1]. Disentangled attention encodes content and position separately, consistently outperforming RoBERTa-base on classification benchmarks.

(2) Focal loss (γ=2). $\mathcal{L} = \frac{1}{N}\sum_i \alpha_{t_i}(1-p_{t_i})^2\,\mathrm{CE}(\log p_i, y_i)$, where $\alpha_1=\sqrt{N_0/N_1} \approx 3\times$ is applied as a separate per-sample scalar (not via `cross_entropy(weight=)`, which double-compensates). The focal factor is *not detached*; detaching creates a degenerate fixed point at $p \approx 0.41$ (verified analytically), causing F_1=0 throughout training.

(3) Prior bias init. Classifier bias initialised to $[0, \log(N_{\mathrm{PCL}}/N_{\mathrm{NoPCL}})] \approx [0, -2.2]$, anchoring $p(\mathrm{PCL}) \approx 0.095$ at step 0 so focal gradients are correctly scaled from the first update.

(4) Layer-wise LR decay (LLRD, λ=0.9). $\mathrm{lr}_k = \mathrm{lr}_{\mathrm{base}} \times 0.9^{L+1-k}$: the head trains at the full 2×10^{-5}; embeddings at $\approx 5\times10^{-6}$, preserving pre-trained representations.

(5) Cosine schedule, keyword-stratified split, in-memory best-model tracking. A 6 % warmup (113 steps) clears the ramp before the epoch-1 checkpoint (step 235), followed by cosine decay. The training set (N=8,839) is split 90/10 stratified by label and keyword. `BestF1Callback` snapshots `model.state_dict()` on each val-F_1 improvement (patience 5) and restores it before evaluation. Training: effective batch 32 (16×grad-accum 2), `max_length=256`, up to 8 epochs.

3.2 Rationale and Expected Outcome

The 9.5 : 1 imbalance draws cross-entropy models toward the majority class; focal loss with $\sqrt{\text{inverse-frequency}}$ weights addresses this, and the prior bias init ensures correct gradient scaling from step 0. DeBERTa-v3-base's disentangled attention better encodes the relational framing of PCL without the overfitting risk of larger variants (Section 1). LLRD protects lower-layer representations critical for this pragmatics-heavy task. Keyword stratification prevents the model from learning community-specific PCL rates rather than genuine condescension. Together these five changes are expected to yield a dev-set F_1 meaningfully above 0.48.

4 Exercise 4: Proposed Approach Implementation (Stage 4)

Repository link on front page.

5 Exercise 5: Evaluation and Error Analysis (Stage 5)

5.1 Global Evaluation

Omitted since results were not as expected; see next section for analysis.

5.2 Local Evaluation

Table 2 summarises performance on the official dev set (2,094 paragraphs).

Model	Precision	Recall	F_1 (PCL)
Baseline (RoBERTa-base, CE)	—	—	0.579
Proposed (DeBERTa-v3-base, Focal)	0.000	0.000	0.000

Table 2: Dev-set results. Baseline exceeded its reported 0.48; proposed model collapsed.

The proposed model degenerated into a constant *No PCL* predictor (90.5 % accuracy, 0 true positives, 199 false negatives). The root cause is that `BestF1Callback` never recorded a positive val F_1, so `restore_best()` was a no-op and the saved checkpoint reflects the fully collapsed state. Likely contributors: (1) the base LR of 2×10^{-5} is too aggressive for DeBERTa-v3-base on a small dataset — the original paper recommends 5–9 $\times 10^{-6}$; (2) the $\sqrt{\text{inverse-frequency}}$ class weight ($\approx 3\times$) may be insufficient to overcome the 9.5 : 1 majority-class gradient pressure at batch size 32; (3) the five changes were not validated jointly before full training.

Given more time the priorities would be: reduce LR to 5×10^{-6}; raise the class-weight exponent to 0.75; add a 2-epoch sanity check monitoring mean p(PCL) before committing to full training; and introduce deviations one at a time to isolate any instability. The baseline result (F_1=0.579 > 0.48 reported) confirms the evaluation pipeline is correct.

References

[1] Pengcheng He, Jianfeng Gao, and Weizhu Chen. Debertav3: Improving deberta using electra-style pre-training with gradient-disentangled embedding sharing. *arXiv preprint arXiv:2111.09543*, 2021.

[2] Yinhan Liu, Myle Ott, Naman Goyal, Jingfei Du, Mandar Joshi, Danqi Chen, Omer Levy, Mike Lewis, Luke Zettlemoyer, and Veselin Stoyanov. Roberta: A robustly optimized bert pretraining approach. *arXiv preprint arXiv:1907.11692*, 2019.

[3] Carla Perez Almendros, Luis Espinosa Anke, and Steven Schockaert. Don't patronize me! an annotated dataset with patronizing and condescending language towards vulnerable communities. In Donia Scott, Nuria Bel, and Chengqing Zong, editors, *Proceedings of the 28th International Conference on Computational Linguistics*, pages 5891–5902, Barcelona, Spain (Online), December 2020. International Committee on Computational Linguistics.